

FWA Fraud Detection

Turning claims data into recovered dollars

A working prototype for a health insurer's Special Investigations Unit — an end-to-end data journey on the Databricks Data Intelligence Platform. (*Synthetic data only.*)

Fraud, waste & abuse is a silent margin leak

- FWA runs an estimated **3–10% of claims spend** (NHCAA / CMS).
- For a plan paying **\$2B in annual claims**, that is **\$60M–\$200M leaking every year**.
- SIU teams triage **reactively**, buried in low-value leads and manual review.
- Every week a case waits, overpayment becomes **unrecoverable**.

The question isn't whether FWA is happening — it's which providers and claims to work first.

The outcome we drive — in your KPIs

KPI	What moves
Recovered / prevented overpayment (\$)	The headline number — dollars back on existing spend, quarter over quarter
SIU productivity (cases / FTE)	More cases closed per investigator; triage in minutes, not weeks
Time-to-triage	Highest-risk providers surface automatically, ranked by recovery potential
Referral precision	Higher share of investigated providers that actually recover

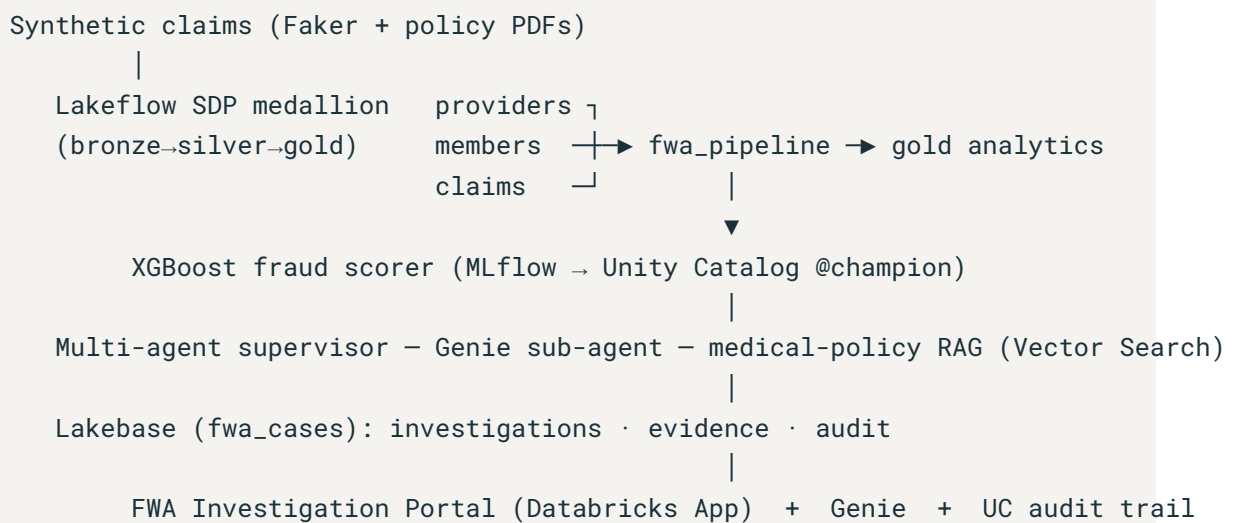
One integrated data journey — not six tools

Stage	Capability
Ingest	

Stage	Capability
	Lakeflow — raw claims into a governed bronze→silver→gold medallion
Govern	Unity Catalog — PHI/PII masking, row filters, lineage, tags
Serve	Lakebase — operational case management (queue, evidence, audit)
Make it intelligent	ML + GenAI — XGBoost fraud scorer <i>and</i> a multi-agent investigator
Query in natural language	Genie — ask the data in plain English
Surface to the business	Databricks App — the FWA Investigation Portal

One platform, one governance model, one copy of the data.

Architecture: raw data to recovered dollars



Everything lands in **one governed Unity Catalog schema** — tables, the model, the RAG index, and the agent trace tables.

What we built — and proof it ran

Captured live from a fully-green end-to-end run (16/16 tasks) on serverless:

Signal	Result
Medical claims ingested	417,489
Pharmacy claims ingested	142,776
FWA signals detected	36,015
Providers risk-ranked	1,488
Claims scored by the model	139,158
Multi-agent evaluation cases (LLM-judged)	24
PHI/PII protections enforced	5 column masks + line-of-business row filter

Execution evidence — row counts, model output, agent evaluation, Genie SQL — is committed as text in the repo's `evidence/` folder.

Value for the executive sponsor

CFO / SIU Director

- **Direct financial recovery** on spend you are already making — not a cost center.
- **Governed and auditable end to end** — PHI/PII masks, full lineage, and captured agent traces stand up to compliance and audit.
- **No new infrastructure** — it runs on the Databricks platform you already govern; nothing new to procure or secure.

Value for the domain owner

SIU Investigator

- **A prioritized, risk-ranked queue** — start each day on the highest-recovery leads, not a spreadsheet.
- **Investigate in plain English** — the agent and Genie pull claims, provider history, ML scores, and medical policy on demand.
- **Evidence and audit trail captured automatically** — every action and rationale is logged for the case file.

Why Databricks

- **One platform** — ingestion, governance, ML, GenAI, and the app on a single lakehouse. No stitching, no data copies.
 - **Governance built in** — PHI/PII masking, lineage, and audit enforced at the platform layer, not bolted on in app code.
 - **AI where the data lives** — the fraud model, the agents, and Genie all read the same governed tables.
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Next steps

- Point the pipeline at a **de-identified extract** of your claims.
- **Calibrate** the fraud model and rules to your denial and billing patterns.
- **Pilot** with your SIU team on a live case backlog.
- **Measure** recovered \$ and time-to-triage against today's baseline.

Let's run a scoped pilot on your data.